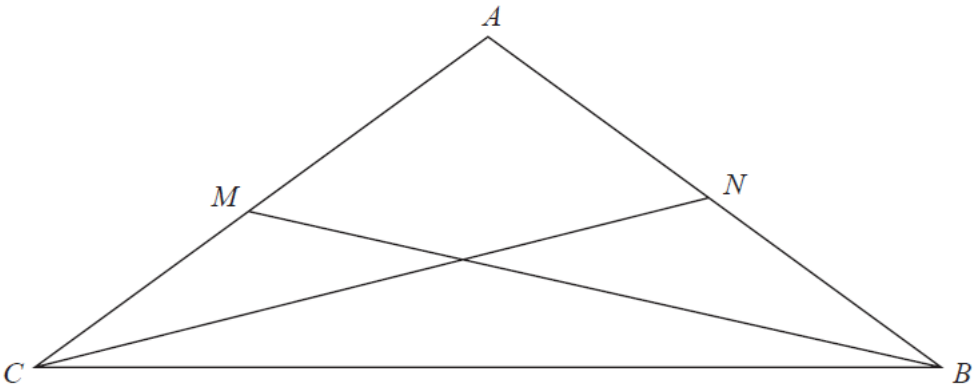
14 The diagram shows an isosceles triangle ABC with $AB = AC$.



M is the midpoint of AC.
N is the midpoint of AB.

Prove that triangle MBC is congruent to triangle NCB.

Angle MCB = angle NBC.

1 mark
Equal included angles.

Base angles of isosceles triangle ABC are equal.

MC = ½AC and NB = ½AB.

Since AC = AB, MC = NB.

1 mark
Equal half-sides.

BC is common to both triangles.

1 mark
Common side; SAS.

Therefore triangles MBC and NCB are congruent by SAS.

(Total for Question 14 is 3 marks)